

ANNEXURE A - PROJECT REGISTRATION AND SYNOPSIS FORM

To be typed on A4 page, signed by the student and the Project Guide, and submitted to the Project Coordinator on or before 26 August 2026.

Field	Entry
1. Name of the Student	Prateek Veeresh Aggimath
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6. Name of the Project Guide	Prof .Sumit Singha Chowdhury
7. Project Title	Scholarship Eligibility Prediction System
8. Domain / Area	Machine Learning, Artificial intelligence, Data science
9. Mode of Work	In-house (Department laboratory)

10. Problem Statement

Educational institutions receive many scholarship applications and manually checking marks, income, Educational institutions receive a large number of scholarship applications every academic year, and manually verifying each student's marks, family income, attendance, and category against eligibility criteria is a time-consuming and labor-intensive task, often leading to delayed decisions, increased administrative burden, evaluation errors, and deserving students being overlooked. Most existing scholarship evaluation systems rely on fixed, rule-based logic and fail to provide a quantifiable eligibility score to support decision-making, and many institutions also lack a dedicated platform for entering student data and analyzing eligibility trends. There is therefore a clear need for a system that can accurately evaluate student information, classify applicants as Eligible or Not Eligible, generate an eligibility score, and present student data through meaningful visualizations—helping institutions identify deserving students efficiently, make informed and consistent decisions, and reduce the manual effort involved in scholarship evaluation.

11. Objectives

No.	Objective	Measure of completion
O1	Scholarship Eligibility Prediction	The goal is to create a system that uses machine learning to analyze student information such as marks, family income, attendance percentage and category criteria to determine whether a student is eligible for a scholarship.
No.	Objective	Measure of completion
O2	Automated Eligibility Prediction and Scoring	The system aims to make the scholarship eligibility checking process easier by automatically predicting whether a student is eligible or not eligible and providing an eligibility score, thereby reducing mistakes and delays caused by manual evaluation.
O3	Web-Based Student Analytics and Visualization	The system aims to develop a website that displays scholarship predictions and student information using charts and graphs. This will help institutions identify deserving students and make fair, transparent and fact-based decisions.

12. Proposed Methodology / Architecture (in brief)

The system follows these main steps:

Student Data Collection → Data Cleaning → Data Preparation → Data Splitting → Model Training & Evaluation → Eligibility Prediction → Flask Web Application → Dashboard & Student Analytics

The methodology uses student data such as marks, family income, attendance, category and scholarship status. Data is cleaned and prepared using Pandas, categorical information is converted into numerical form, the dataset is divided into training and testing data, and classification models are trained and evaluated using scikit-learn. The selected model predicts scholarship eligibility and provides an eligibility score/probability. A Flask web application displays the results and analytics through charts and graphs.

13. Tools, Technology Stack, Dataset and Hardware

Category	Details
Application stack	Python + Flask (web application); HTML + CSS + Bootstrap + JavaScript (client)
Database	MySQL
ML / Analysis	Scikit-learn, Pandas, NumPy, Matplotlib, Plotly
API	Logistic Regression, Decision Tree, Random Forest, Support Vector Machine
Version control	GitHub
Dataset	Student data including marks, family income, attendance percentage, category, scholarship status and other relevant details
Hardware	Personal laptop / computer

14. Expected Deliverables and Intended Outcome

- Scholarship Eligibility Prediction web portal.
- Prediction of **Eligible** / **Not Eligible** students.
- Eligibility score/probability.
- Student analytics and statistics.
- Charts and graphs showing eligibility distribution, academic performance, income-level analysis and other relevant patterns.
- Faster and easier scholarship eligibility evaluation.
- Reduced manual effort and errors.
- More transparent and fair scholarship decision-making.

Student

Prateek Veeresh Aggimath

Project Guide

Prof .Sumit Singha Chowdhury

Project Coordinator

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